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B.Tech Artificial Intelligence & Data Science - 2nd Semester

Object Oriented Programming – Lab Assignment 10

Program 1: Creating and Using a User-Defined Package

Calculator.java :

```
package mypackage;
public class Calculator{
    public int add(int a, int b){
        return a+b;
    }
    public int sub(int a, int b){
        return a-b;
    }
}
```

MainApp.java :

```
import mypackage.Calculator;
public class MainApp{
    public static void main(String[] args){
        Calculator obj = new Calculator();
        System.out.println("Addition: "+obj.add(10,5));
        System.out.println("Subtraction: "+obj.sub(10,5));
    }
}
```

Output:

```
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10$ cd src/mypackage/
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/mypackage$ javac -d . Calculator.java
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/mypackage$ cd ..
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ javac MainApp.java
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ java MainApp
Addition: 15
Subtraction: 5
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$
```

Program 2: Using Fully Qualified Name (Without import)

Demo.java :

```
public class Demo{  
    public static void main(String[] args) {  
        mypackage.Calculator obj = new mypackage.Calculator();  
        System.out.println("Addition : "+obj.add(7,3));  
    }  
}
```

Output:

```
/usr/java/jdk-17-oracle-x64/bin/java -javaagent:/home/krish/.local/share/  
Addition : 10  
  
Process finished with exit code 0
```

Program 3: Creating and Using a Sub-Package

Message.java :

```
package com.company.utils;
public class Message {
    public void display(){
        System.out.println("Hello from subpackage");
    }
}
```

Main.java :

```
import com.company.utils.Message;
public class Main {
    public static void main(String[] args) {
        Message msg = new Message();
        msg.display();
    }
}
```

Output:

```
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ cd com/company/utils/
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/com/company/utils$ javac -d . Message.java
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/com/company/utils$ pwd
/home/krish/Desktop/Java-Archive/00P/Lab 10/src/com/company/utils
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/com/company/utils$ cd ..
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/com/company$ cd ..
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src/com$ cd ..
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ javac Main.java
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ java Main.java
Hello from subpackage
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$
```

Program 4: Accessing Classes from Another Package

A.java :

```
package pack1;
public class A{
    public void show(){
        System.out.println("Inside Package Pack1");
    }
}
```

B.java :

```
package pack2;
import pack1.A;
public class B{
    public static void main(String[] args){
        A obj = new A();
        obj.show();
    }
}
```

Output:

```
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ javac -d . pack1/A.java pack2/B.java
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$ java pack2.B
Inside Package Pack1
(base) krish@krish-Predator-PHN16-71:~/Desktop/Java-Archive/00P/Lab 10/src$
```